

Charito

Effective scholarly tools for layman biblical study

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The Problem

- Scholarly tools for biblical research are typically gated behind expensive applications like Logos or Accordance.
- The raw resources for those tools are typically public domain.

The Solution

- Charito uses public domain resources to provide tools that enhance in-depth study of scripture.
- Valuable documents are available from the millennia of church history.
- It does so via a freely available web application.

The Resources

- The Christian Classics Ethereal Library provides historical documents as XML files.
- The CCEL provides these as PDF files but doesn't put them into a comprehensive Bible application with the other sources I include.
- Other sources provide Greek and Hebrew texts via XML.
- Public-domain bibles like the KJV are also available as XML.

Constraints

It Must Be:

- Stable
- Efficient
- Maintainable by single person

The Languages

- Common Lisp: Powerful programming language for creating Domain Specific Languages
- AP5: Advanced in-memory database represented by a logic-programming language.
- Spinneret: Write HTML in Lisp
- Parenscript: Write JavaScript in Lisp

Lisp vs Python/PHP/Ruby/Perl

Common Lisp	Scripting Language
Compiles to efficient machine code	Bytecode-interpreted or JIT
Strong type enforcement at compile time and run time	No compile-time or run-time type enforcement
Macros allow for easy creation of efficient DSLs by operating on the AST at compile-time	The compiler only compiles the base language with no comparable extension
Conditions & restarts for error-handling	Try/catch or function return value only

The Tools

- SBCL Lisp Compiler: Stable and efficient
- Plump: Lenient XML parser
- Woo: Eitaro Fukamachi's blazing-fast web server

Testing

```
* (asdf:test-system :jweb)

Running test suite JWEB-TEST
Running test suite FRAMEWORK
Running test MAKE-ID-STRING .
Running test APPLY-MIDDLEWARE-BASIC .
Running test APPLY-MIDDLEWARE-CHAIN ..
Running test suite REF-TREE
Running test SINGLE-VERSE ..
Running test MULTI-VERSE ..
Running test MULTI-CHAPTER ..
Did 10 checks.
Pass: 10 (100%)
Skip: 0 ( 0%)
Fail: 0 ( 0%)

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* █
```

Demonstration

- I shall now demonstrate how the application functions

Conclusion

- This rapidly-evolving application will continue to integrate more resources into the database and create tools with those resources
- Roadmap: After integrating lexicons, tools like flashcards and parsings will begin to be available. More ideas available as users request.